7. Symmetric Matrices and Quadratic Forms

7.1 Diagonalization of Symmetric Matrices

Theorem 1

If A is symmetric, then any two eigenvectors from different eigenvalues are orthogonal.

Definition: Orthogonally Diagonalizable

A $n \times n$ matrix A is orthogonally diagonalizable if there exists an orthogonal matrix P and a diagonal matrix D such that $A = PDP^{-1} = PDP^{\top}$ (since P is orthogonal, $P^{-1} = P^{\top}$).

Theorem 2

A $n \times n$ matrix A is orthogonally diagonalizable if and only if A is a symmetric matrix.

Theorem 3: The Spectral Theorem for Symmetric Matrices

An $n \times n$ symmetric matrix A has the following properties:
a. A has n real eigenvalues, counting multiplicities.
b. The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.
c. The eigenspaces are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal.
d. A is orthogonally diagonalizable.

How to Find P?

1. Find the eigenvectors of A.
2. Eigenvectors corresponding to different eigenvalues are orthogonal (by Theorem 3c).
3. Normalize all eigenvectors to get orthonormal eigenvectors $\vec{u}_1, \cdots, \vec{u}_n$.

Spectral Decomposition of a Matrix

Suppose $A = PDP^{-1}$, where the columns of P are orthonormal eigenvectors $\vec{u}_1, \ldots, \vec{u}_n$ of A and the corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$ are in the diagonal matrix D. Then, since $P^{-1} = P^\top$,

$$\begin{aligned} A = PDP^\top &= (\vec{u}_1 \ \ldots \ \vec{u}_n) \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} \vec{u}_1^\top \\ \vdots \\ \vec{u}_n^\top \end{pmatrix} \\ &= (\lambda_1 \vec{u}_1 \ \ldots \ \lambda_n \vec{u}_n) \begin{pmatrix} \vec{u}_1^\top \\ \vdots \\ \vec{u}_n^\top \end{pmatrix} \\ &= \lambda_1 \vec{u}_1 \vec{u}_1^\top + \cdots + \lambda_n \vec{u}_n \vec{u}_n^\top \end{aligned}$$

This is called a spectral decomposition of A.

7.2 Quadratic Forms

Definition: Quadratic Forms

A quadratic form on $\mathbb{R}^n$ is a function Q defined on $\mathbb{R}^n$ whose value at a vector $\vec{x}$ in $\mathbb{R}^n$ is computed by $Q(\vec{x}) = \vec{x}^\top A \vec{x}$, where A is an $n \times n$ symmetric matrix (called the matrix of the quadratic form).

Change of Variable in a Quadratic Form

If $\vec{x}$ represents a vector in $\mathbb{R}^n$, then a change of variable is an equation of the form $\vec{x} = P\vec{y}$ (or $\vec{y} = P^{-1}\vec{x}$), where P is an invertible matrix and $\vec{y}$ is a new "variable vector". The columns of P determine the new coordinate system.

For a quadratic form $Q(\vec{x}) = \vec{x}^\top A \vec{x}$, substituting $\vec{x} = P\vec{y}$ gives:

$$\vec{x}^\top A \vec{x} = (P\vec{y})^\top AP\vec{y} = \vec{y}^\top P^\top AP\vec{y} = \vec{y}^\top (P^\top AP)\vec{y}$$

The new matrix of the quadratic form is $P^\top AP$. Since A is symmetric, Theorem 2 guarantees that there exists an orthogonal matrix P such that $P^\top AP$ is a diagonal matrix D, and the new quadratic form is $\vec{y}^\top D\vec{y}$ (with no cross-product terms).

Theorem 4: The Principal Axes Theorem

Let A be an $n \times n$ symmetric matrix. Then there is an orthogonal change of variable $\vec{x} = P\vec{y}$ that transforms the quadratic form $\vec{x}^\top A\vec{x}$ into a quadratic form $\vec{y}^\top D\vec{y}$ with no cross-product terms.

Definition: Definiteness of Quadratic Forms

A quadratic form Q is:
a. **positive semidefinite** if $Q(\vec{x}) \geq 0$ for all $\vec{x} \neq \vec{0}$;
b. **negative definite** if $Q(\vec{x}) < 0$ for all $\vec{x} \neq \vec{0}$;
c. **negative semidefinite** if $Q(\vec{x}) \leq 0$ for all $\vec{x} \neq \vec{0}$;
d. **indefinite** if $Q(\vec{x})$ assumes both positive and negative values.

Theorem 5: Quadratic Forms and Eigenvalues

Let A be an $n \times n$ symmetric matrix. Then the quadratic form $\vec{x}^\top A\vec{x}$ is:
a. positive definite if and only if all eigenvalues of A are positive;
b. negative definite if and only if all eigenvalues of A are negative;
c. indefinite if and only if A has both positive and negative eigenvalues.

Definitions of Definite Matrices

- A **positive definite matrix** A is a symmetric matrix for which the quadratic form $\vec{x}^\top A\vec{x}$ is positive definite.
- A **positive semidefinite matrix** is defined analogously, as are negative definite and negative semidefinite matrices.

Supplementary Properties of Symmetric Positive Definite Matrices

1. If S and T are symmetric positive definite matrices, then $S + T$ is also a symmetric positive definite matrix.
2. A symmetric matrix S is positive definite if and only if it satisfies any (and hence all) of the following properties:
   (1) All leading principal minors of S are positive;
   (2) All pivots of S are positive;
   (3) All n eigenvalues of S are positive;
   (4) $\vec{x}^\top S\vec{x} > 0$ for all $\vec{x} \neq \vec{0}$;
   (5) $S = A^\top A$ where the columns of A are linearly independent.

Inertia Theorem

Let the rank of the quadratic form $f = \vec{x}^\top A\vec{x}$ be r. If there exist two invertible transformations $\vec{x} = Cy$ and $\vec{x} = Pz$ such that:

$$f = k_1 y_1^2 + k_2 y_2^2 + \cdots + k_r y_r^2 \quad (k_i \neq 0)$$

and

$$f = \lambda_1 z_1^2 + \lambda_2 z_2^2 + \cdots + \lambda_r z_r^2 \quad (\lambda_i \neq 0)$$

then the number of positive values among k_i is equal to the number of positive values among λ_i, and the number of negative values among k_i is equal to the number of negative values among λ_i.

- The number of positive values is called the **positive inertia index**;
- The number of negative values is called the **negative inertia index**.

7.3 Constrained Optimization

Theorem 6

Let A be a symmetric matrix, and define $m = \min\{\vec{x}^\top A\vec{x} \mid \vec{x}^\top \vec{x} = 1\}$ and $M = \max\{\vec{x}^\top A\vec{x} \mid \vec{x}^\top \vec{x} = 1\}$. Then:

- M is the greatest eigenvalue of A, achieved when $\vec{x}$ is a unit eigenvector corresponding to M;
- m is the least eigenvalue of A, achieved when $\vec{x}$ is a unit eigenvector corresponding to m.

Theorem 7

Let A be a symmetric matrix, and let $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ be its eigenvalues. Then the minimum value of $\vec{x}^\top A\vec{x}$ subject to the constraints $\vec{x}^\top \vec{x} = 1$ and $\vec{x}^\top \vec{u}_1 = 0$ (where $\vec{u}_1$ is a unit eigenvector corresponding to λ_1) is λ_2, achieved when $\vec{x}$ is a unit eigenvector $\vec{u}_2$ corresponding to λ_2.

Theorem 8

Let A be an $n \times n$ symmetric matrix with orthogonal diagonalization $A = PDP^\top$, where the diagonal entries of D are arranged as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ and the columns of P are the corresponding unit eigenvectors $\vec{u}_1, \vec{u}_2, \cdots, \vec{u}_n$. Then for each $k = 1, 2, \cdots, n$, the maximum value of $\vec{x}^\top A\vec{x}$ subject to the constraints:

$$\vec{x}^\top \vec{x} = 1, \quad \vec{x}^\top \vec{u}_1 = 0, \quad \cdots, \quad \vec{x}^\top \vec{u}_{k-1} = 0$$

is λ_k, achieved when $\vec{x} = \vec{u}_k$.

7.4 Singular Value Decomposition (SVD)

Definition: Singular Values

The **singular values** of an $m \times n$ matrix A are the square roots of the eigenvalues of $A^\top A$, denoted by $\sigma_1, \sigma_2, \cdots, \sigma_n$, and arranged in decreasing order: $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, 2, \cdots, n$, where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ are the eigenvalues of $A^\top A$.

Singular Values as Lengths of $A\vec{v}_i$

The singular values of A are the lengths of the vectors $A\vec{v}_1 \ldots A\vec{v}_n$.

Proof

Suppose the eigenvectors of $A^\top A$ are $\vec{v}_1 \ldots \vec{v}_n$.

First, compute the squared norm of $A\vec{v}_i$:

$$\|A\vec{v}_i\|^2 = (A\vec{v}_i)^\top A\vec{v}_i = \vec{v}_i^\top A^\top A\vec{v}_i$$

Since $\vec{v}_i$ is an eigenvector of $A^\top A$, $A^\top A\vec{v}_i = \lambda_i \vec{v}_i$, so:

$$\|A\vec{v}_i\|^2 = \vec{v}_i^\top \lambda_i \vec{v}_i = \lambda_i \vec{v}_i^\top \vec{v}_i$$

Then suppose $\{\vec{v}_1 \ldots \vec{v}_n\}$ is an **orthonormal basis** (so $\vec{v}_i^\top \vec{v}_i = 1$), thus:

$$\|A\vec{v}_i\|^2 = \lambda_i$$

Note (P.S.)

$\vec{v}_i$为特征向量构成的标准正交基。

Theorem 9

Suppose $\{\vec{v}_1, \vec{v}_2, \cdots, \vec{v}_n\}$ is an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of $A^\top A$, arranged so that the corresponding eigenvalues satisfy $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Suppose A has r

nonzero singular values (i.e., $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ and $\sigma_{r+1} = \cdots = \sigma_n = 0$). Then:

- $\{A\vec{v}_1, A\vec{v}_2, \cdots, A\vec{v}_r\}$ is an orthogonal basis for $\text{Col } A$;
- $\text{rank } A = r$.

Definition: Σ

The decomposition of A involves an $m \times n$ "diagonal" matrix Σ:

$$\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$$

where D is an $r \times r$ diagonal matrix ($r \leq \min\{m, n\}$)

Theorem 10: The Singular Value Decomposition (SVD)

Let A be an $m \times n$ matrix with rank r. Then there exists:

- An $m \times m$ orthogonal matrix U (left singular vectors);
- An $n \times n$ orthogonal matrix V (right singular vectors);
- A diagonal matrix Σ of size $m \times n$, where the diagonal entries are the first r singular values of A : $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$, and the remaining entries are 0;

such that:

$$A = U \Sigma V^\top$$

PROOF

Let λ_i and $\vec{v}_i$ be as in Theorem 9, so that $\{A\vec{v}_1, \ldots, A\vec{v}_r\}$ is an orthogonal basis for $\text{Col } A$.

Normalize each $A\vec{v}_i$ to obtain an orthonormal basis $\{\vec{u}_1 \ldots \vec{u}_r\}$:

$$\vec{u}_i = \frac{1}{\|A\vec{v}_i\|} A\vec{v}_i = \frac{1}{\sigma_i} A\vec{v}_i$$

and

$$A\vec{v}_i = \sigma_i \vec{u}_i \quad (1 \leq i \leq r)$$

Now extend $\{\vec{u}_1 \ldots \vec{u}_r\}$ to an orthonormal basis $\{\vec{u}_1 \ldots \vec{u}_m\}$ of $\mathbb{R}^m$, and let:

$$U = (\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_m), \quad V = (\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n)$$

And then:

$$AV = [A\vec{v}_1 \ \dots \ A\vec{v}_r \ 0 \ \dots \ 0] = [\sigma_1\vec{u}_1 \ \dots \ \sigma_r\vec{u}_r \ 0 \ \dots \ 0]$$

$$U\Sigma = [\vec{u}_1 \ \vec{u}_2 \ \dots \ \vec{u}_r \ \dots \ \vec{u}_m] \begin{bmatrix} \sigma_1 & & & & 0 \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_r & \\ & & & & 0 \end{bmatrix}$$

$$= (\sigma_1\vec{u}_1 \ \sigma_2\vec{u}_2 \ \dots \ \sigma_r\vec{u}_r \ 0 \ \dots \ 0)$$

So

$$AV = U\Sigma$$

And since V is an orthogonal matrix:

$$A = U\Sigma V^\top$$

Theorem: The Invertible Matrix Theorem (concluded)

Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix:

- u. $(\text{Col } A)^\perp = \{0\}$
- v. $(\text{Nul } A)^\perp = \mathbb{R}^n$
- w. $\text{Row } A = \mathbb{R}^n$
- x. A has n nonzero singular values.

Theorem 11: Orthogonal Bases from SVD

Given an SVD $A = U\Sigma V^\top$ of an $m \times n$ matrix A, let $r = \text{rank } A$. Then:

- $\{\vec{u}_1, \cdots, \vec{u}_r\}$ is an orthonormal basis for $\text{Col } A$;
- $\{\vec{u}_{r+1}, \cdots, \vec{u}_m\}$ is an orthonormal basis for $\text{Null } A^\top$ (left null space);
- $\{\vec{v}_1, \cdots, \vec{v}_r\}$ is an orthonormal basis for $\text{Row } A$;
- $\{\vec{v}_{r+1}, \cdots, \vec{v}_n\}$ is an orthonormal basis for $\text{Null } A$ (null space).

Reduced Singular Value Decomposition

Let $U_r = [\vec{u}_1 \cdots \vec{u}_r]$ ($m \times r$), $V_r = [\vec{v}_1 \cdots \vec{v}_r]$ ($n \times r$), and $D = \text{diag}(\sigma_1, \cdots, \sigma_r)$ ($r \times r$ diagonal matrix). Then the reduced SVD of A is:

$$A = U_r D V_r^\top$$

Moore-Penrose Inverse (Pseudoinverse)

The Moore-Penrose inverse of A, denoted A^+, is defined as:

$$A^+ = V_r D^{-1} U_r^\top$$

where $D^{-1} = \mathrm{diag}(1/\sigma_1, \cdots, 1/\sigma_r)$.

补充：如何证明A^TA的特征值为V中向量

证 明:
设$A = U\Sigma V^T$（SVD分解形式）
则

$$\begin{aligned} A^TA &= (U\Sigma V^T)^T(U\Sigma V^T) \\ &= V\Sigma^T U^T U\Sigma V^T \quad (\text{因 } U \text{为正交矩阵，故 } U^TU = I) \\ &= V\Sigma^T\Sigma V^T \end{aligned}$$

由于A^TA为对称矩阵（故可对角化），且$\Sigma^T\Sigma$为对角矩阵，因此$V(\Sigma^T\Sigma)V^T$是A^TA的对角化形式。

SVD分解 流程

1. 计算A^TA的特征值与特征向量：
   特征值：$\lambda_1, \lambda_2, \ldots, \lambda_r$
   特征向量：$\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_r$
2. 得到奇异值：

$$\sigma_1 = \sqrt{\lambda_1},\ \sigma_2 = \sqrt{\lambda_2},\ \ldots,\ \sigma_r = \sqrt{\lambda_r}$$

3. 计算A的零空间，得到$\vec{v}_{r+1}, \ldots, \vec{v}_n$
4. 计算$\vec{u}_i = \frac{A\vec{v}_i}{\sigma_i}\ (i = 1, 2, \ldots, r)$，得到$\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_r$
5. 计算A的左零空间，得到$\vec{u}_{r+1}, \ldots, \vec{u}_m$
6. 正交化后，A的SVD形式为：

$$A = \begin{pmatrix} \vec{u}_1 & \vec{u}_2 & \dots & \vec{u}_r & \vec{u}_{r+1} & \dots & \vec{u}_m \end{pmatrix} \begin{pmatrix} \sigma_1 & & & & 0 \\ & \ddots & & & \\ & & \sigma_r & & \\ & & & 0 & \\ & & & & \ddots \end{pmatrix} \begin{pmatrix} \vec{v}_1^T \\ \vdots \\ \vec{v}_r^T \\ \vec{v}_{r+1}^T \\ \vdots \\ \vec{v}_n^T \end{pmatrix}$$